

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE) ON LOAN APPROVAL PREDICTION USING MACHINE LEARNING

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING



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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “Loan Approval Prediction Using Machine Learning Algorithms” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of the respective Faculty Mentor.

The matter presented in this project report has not been submitted in any other university/ institute for the award of any degree.

Place: Rajpura

Date: May 17, 2026

(Manan Jain)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

The rapid digital transformation of the banking and financial sector has significantly increased the number of loan applications processed by financial institutions. Traditional loan approval systems rely heavily on manual verification and rule-based decision-making, which are often time-consuming, inconsistent, and prone to human bias.

This project presents a comparative study of three supervised machine learning algorithms—Logistic Regression, K-Nearest Neighbors (KNN), and Naive Bayes—for predicting loan approval outcomes. The study utilizes a loan prediction dataset containing applicant demographic and financial attributes such as applicant income, education, credit history, property area, and loan amount.

Various preprocessing techniques including handling missing values, encoding categorical variables, feature scaling, and train-test splitting were applied to improve model performance. The models were evaluated using standard classification metrics including accuracy, precision, recall, and F1-score.

Experimental results indicate that Logistic Regression achieved the best overall performance due to its balanced accuracy, interpretability, and suitability for binary classification problems. KNN effectively captured non-linear relationships among features but suffered from higher computational complexity, while Naive Bayes provided fast predictions with comparatively lower accuracy because of its feature independence assumption.

The study demonstrates that machine learning techniques can significantly improve the efficiency and reliability of loan approval systems by reducing processing time, minimizing bias, and enabling scalable financial decision-making.

1. Introduction

The banking and financial services industry has experienced rapid technological growth due to advancements in artificial intelligence, data analytics, and machine learning. One of the most critical operations performed by financial institutions is the loan approval process, where banks determine whether an applicant is eligible for credit based on financial and demographic factors.

Traditional loan approval systems depend heavily on manual verification and predefined decision-making rules. Although these methods provide administrative control, they are often slow, inconsistent, and inefficient when processing large numbers of applications. Additionally, manual decision-making introduces the possibility of human bias and errors.

Machine learning has emerged as an effective solution for automating financial decision-making systems. Machine learning algorithms can identify hidden patterns in historical data and use them to predict outcomes for new loan applicants. Several studies have shown that machine learning models outperform conventional statistical techniques in loan approval prediction and credit scoring applications.

This project focuses on comparing three machine learning algorithms—Logistic Regression, K-Nearest Neighbors (KNN), and Naive Bayes—to identify the most suitable model for real-world financial applications.

1.1 Problem Statement

Loan approval is a critical process because incorrect lending decisions can result in financial losses for banks and financial institutions. Approving loans for high-risk applicants increases the possibility of loan defaults, whereas rejecting eligible customers may result in the loss of business opportunities.

Traditional systems struggle to efficiently process growing volumes of loan applications and often fail to handle complex relationships among applicant attributes. Therefore, there is a need for intelligent systems capable of improving loan approval accuracy and efficiency.

1.2 Objectives

The main objectives of this project are:

- To implement Logistic Regression, KNN, and Naive Bayes algorithms for loan prediction
- To preprocess the dataset using suitable data cleaning techniques
- To evaluate the performance of models using accuracy, precision, recall, and F1-score
- To compare the strengths and limitations of each algorithm
- To identify the most suitable machine learning model for practical financial applications

2. METHODOLOGY

The methodology adopted in this project consists of several stages including data collection, preprocessing, model training, and performance evaluation

2.1 Dataset Collection

The dataset used in this study contains historical loan application records with applicant demographic and financial information. Each record represents an individual applicant and includes attributes such as income, education, credit history, loan amount, loan term, and property area.

2.2 Data Preprocessing

Data preprocessing is an essential step in machine learning because raw datasets often contain missing values and inconsistent formats.

Handling Missing Values

Missing values were handled using mean and mode imputation techniques to improve data quality.

Encoding Categorical Variables

Categorical features were converted into numerical form using Label Encoding and One-Hot Encoding methods.

Feature Scaling

Feature scaling was applied to normalize numerical attributes, especially for KNN, which relies on distance calculations.

Train-Test Split

The dataset was divided into training and testing sets using an 80:20 ratio to evaluate model performance effectively and reduce overfitting.

2.3 Machine Learning Model

Logistic Regression

Logistic Regression is a supervised learning algorithm used for binary classification problems. It provides high interpretability and balanced performance.

K-Nearest Neighbors (KNN)

KNN is a distance-based algorithm that classifies data points based on the nearest neighboring samples. It performs well for non-linear datasets but has higher computational complexity.

Naive Bayes

Naive Bayes is a probabilistic classification algorithm based on Bayes' theorem. It is computationally efficient and suitable for large datasets.

3. TOOLS AND TECHNOLOGIES

The following tools and technologies were used during the implementation of this project:

- **Programming Language:** Python
- **Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- **Machine Learning Algorithms:** Logistic Regression, KNN, Naive Bayes
- **Development Environment:** Jupyter Notebook
- **Dataset Processing Techniques:** Label Encoding, Feature Scaling, Missing Value Handling

4. Implementation

The implementation phase involved training and testing the selected machine learning algorithms using the preprocessed dataset.

The following steps were performed during implementation:

1. Importing the dataset
2. Data cleaning and preprocessing
3. Splitting the dataset into training and testing sets
4. Training Logistic Regression, KNN, and Naive Bayes models
5. Evaluating models using classification metrics

The models were tested using accuracy, precision, recall, and F1-score to compare their effectiveness in predicting loan approval decisions.

5. Major Findings/Outcomes/Output/Results

The experimental results demonstrated varying performance levels among the evaluated algorithms.

Algorithm	Accuracy	Precision	Recall	F1-Score
Logistic Regression	High	Balanced	High	High
KNN	Moderate	Good	Moderate	Moderate
Naive Bayes	Moderate	Moderate	Good	Moderate

Key Observations

- Logistic Regression achieved the best overall performance due to its balanced accuracy and interpretability.
- KNN successfully captured non-linear relationships among features but required higher computational resources.
- Naive Bayes provided fast predictions but showed comparatively lower accuracy because of the feature independence assumption.

The results indicate that machine learning algorithms can significantly improve the efficiency and reliability of loan approval systems.

6. Conclusion and Future Scope

This project demonstrates the effectiveness of machine learning algorithms in improving loan approval prediction systems. A comparative analysis of Logistic Regression, KNN, and Naive Bayes was conducted using a loan prediction dataset.

Among the evaluated models, Logistic Regression emerged as the most suitable algorithm due to its high accuracy, balanced performance, and interpretability. KNN captured complex relationships among features, while Naive Bayes offered computational efficiency.

The findings highlight that machine learning techniques can reduce processing time, minimize bias, and improve decision-making in banking systems.

Future Scope

Future work can focus on implementing advanced machine learning algorithms such as:

- Random Forest
- XGBoost
- Support Vector Machines (SVM)
- Deep Learning Models

Additional improvements may include real-time deployment, larger datasets, and integration with cloud-based banking systems.

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